

NUTRIPLAN CAMPUS

School Nutrition Management System

Project Documentation

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Abstract

NutriPlan Campus is a mobile-based School Nutrition Management System developed using Flutter and Firebase technologies. The application helps educational institutions manage student nutrition records, dietary requirements, meal planning, and nutrition reporting.

The system provides secure user authentication, student registration, dietary category management, real-time cloud storage, and reporting features. By replacing manual record keeping with a digital solution, NutriPlan Campus improves efficiency, accuracy, and accessibility of nutrition-related information.

Chapter 1: Introduction

Background of the Study

Nutrition management is an important aspect of student health and well-being. Educational institutions often face challenges in tracking students with special dietary requirements and maintaining accurate nutrition records.

Traditional paper-based systems are difficult to manage, prone to errors, and inefficient when handling large numbers of students.

The advancement of mobile technologies provides opportunities to develop systems that simplify nutrition management while ensuring data accuracy and accessibility.

NutriPlan Campus was developed to address these challenges.

Problem Statement

Many schools face challenges such as:

- Poor management of student nutrition records.
- Difficulty tracking dietary requirements.
- Lack of centralized student information.
- Time-consuming manual record keeping.
- Limited reporting capabilities.

These challenges reduce efficiency and increase the possibility of errors.

Proposed Solution

NutriPlan Campus provides a mobile application that enables institutions to:

- Register and manage students.
- Manage dietary categories.
- Store records securely in the cloud.
- Generate nutrition reports.
- Improve nutrition management efficiency.

Chapter 2: Project Objectives

Main Objective

To develop a mobile application that improves student nutrition management within educational institutions.

Specific Objectives

- To provide secure user authentication.
- To manage student records digitally.
- To categorize students according to dietary requirements.
- To provide cloud-based storage using Firebase.
- To generate nutrition reports and statistics.
- To provide an easy-to-use mobile interface.

Chapter 3: Scope of the Project

The project focuses on student nutrition management activities.

Included

- User registration

- User login
- Student management
- Diet management
- Search functionality
- Cloud database integration
- Reporting and statistics

Excluded

- Online food ordering
- Payment processing
- Student billing
- Food delivery services
- Supplier management

Chapter 4: System Analysis

Existing System

Many institutions still maintain nutrition records manually using notebooks, spreadsheets, or paper forms.

Limitations

- Time-consuming processes.
- Data duplication.
- Human errors.
- Difficulty generating reports.
- Poor accessibility of information.

Proposed System

NutriPlan Campus digitizes nutrition management through a mobile application connected to a cloud database.

The system allows authorized users to manage student information and dietary records efficiently.

Chapter 5: System Design

Functional Requirements

The system shall:

1. Allow users to create accounts.
2. Allow users to log in securely.
3. Add student records.
4. Edit student information.
5. Delete student records.
6. Search student records.
7. Generate nutrition reports.
8. Store information in the cloud database.

Non-Functional Requirements

- Reliability
- Security
- Usability
- Performance
- Maintainability

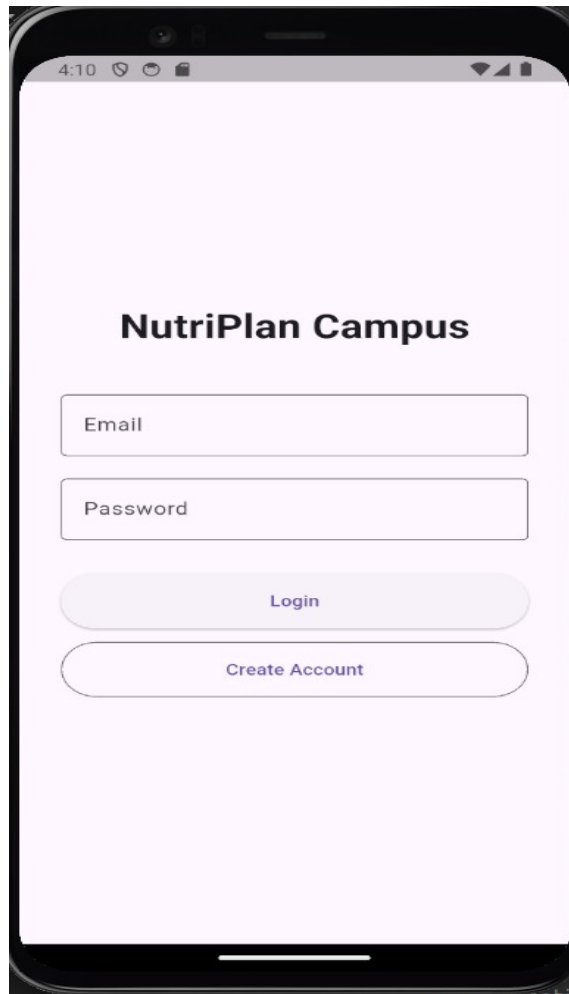
Chapter 6: Technologies Used

Technology	Purpose
Flutter	Mobile application development
Dart	Programming language
Firebase Authentication	User authentication
Cloud Firestore	Cloud database
VS Code	Development environment
GitHub	Version control and project hosting
Android SDK	Testing and deployment

Chapter 7: System Features

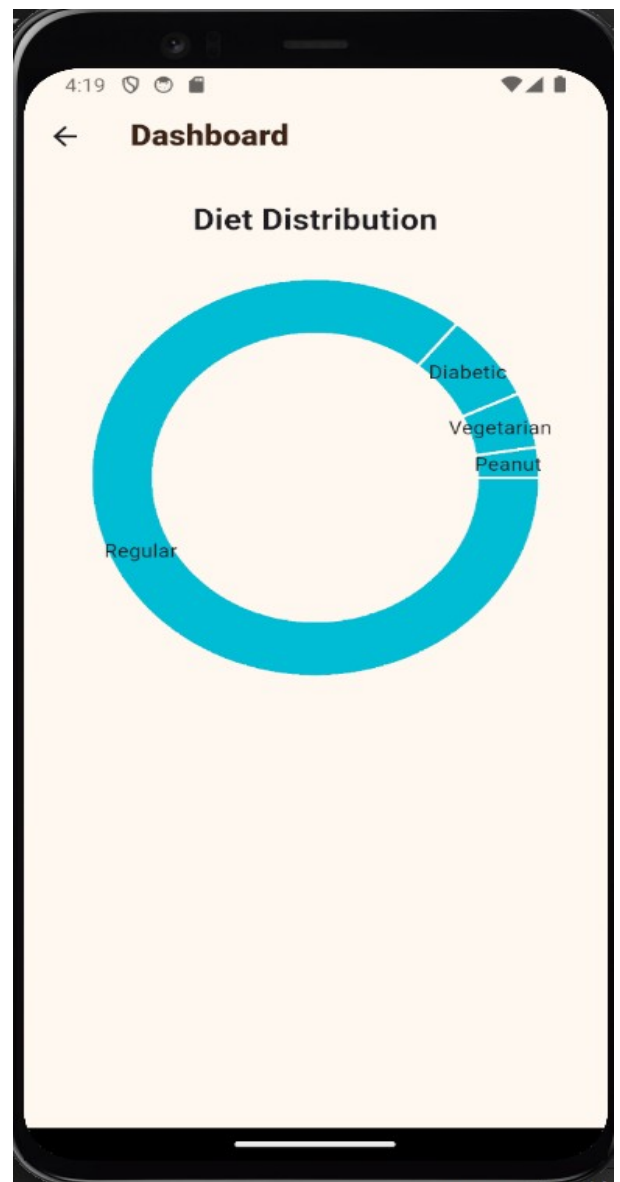
Login and Authentication

The system allows users to register and log in securely using Firebase Authentication.



Dashboard

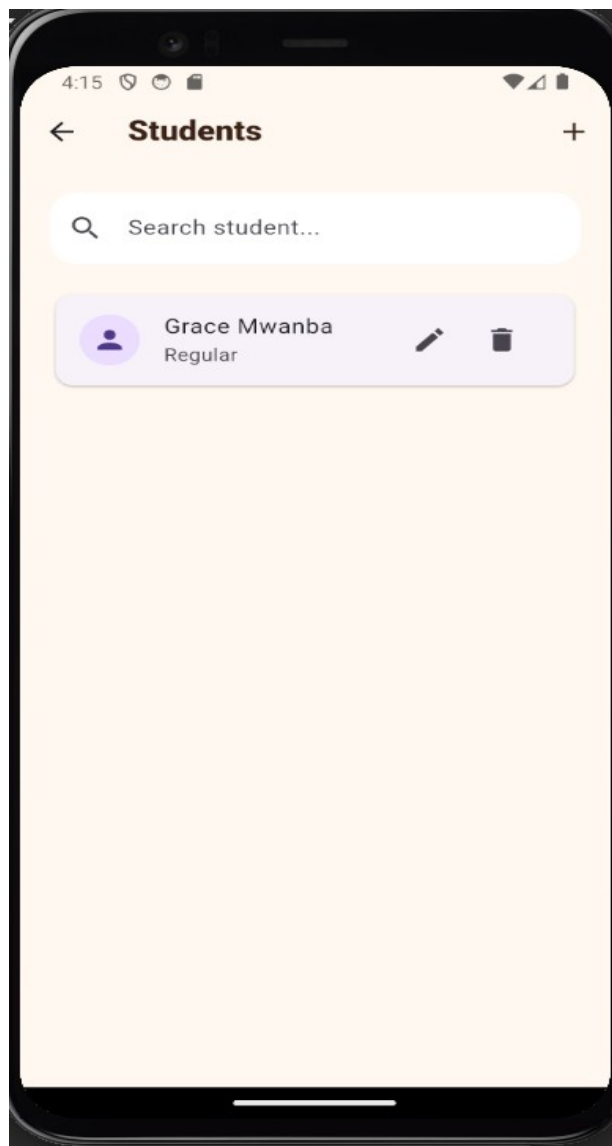
The dashboard provides access to all system modules.



Student Management

Users can:

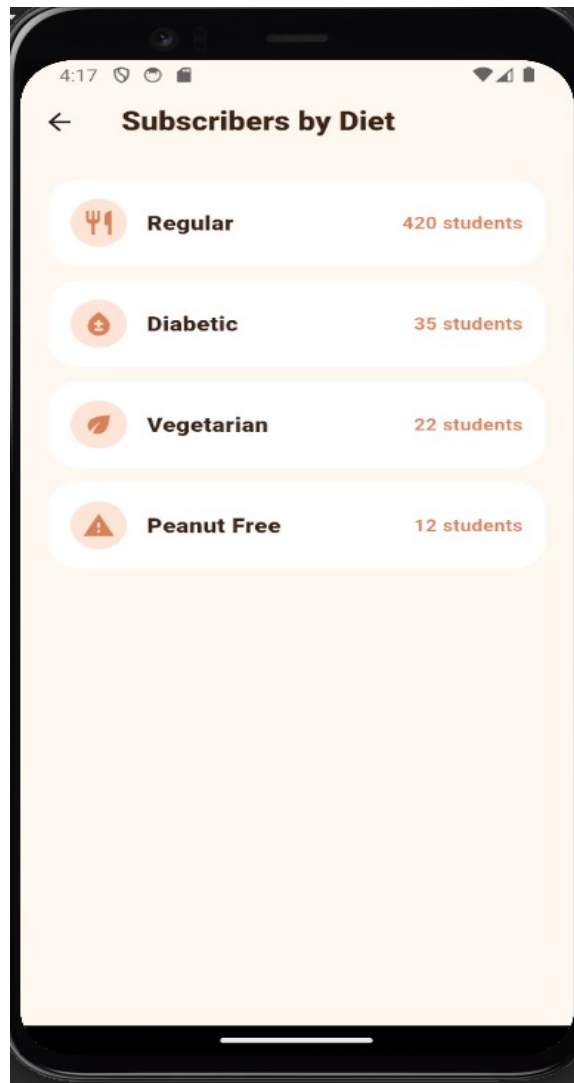
- Add students
- Edit student information
- Delete students
- Search students



Diet Management

The system supports:

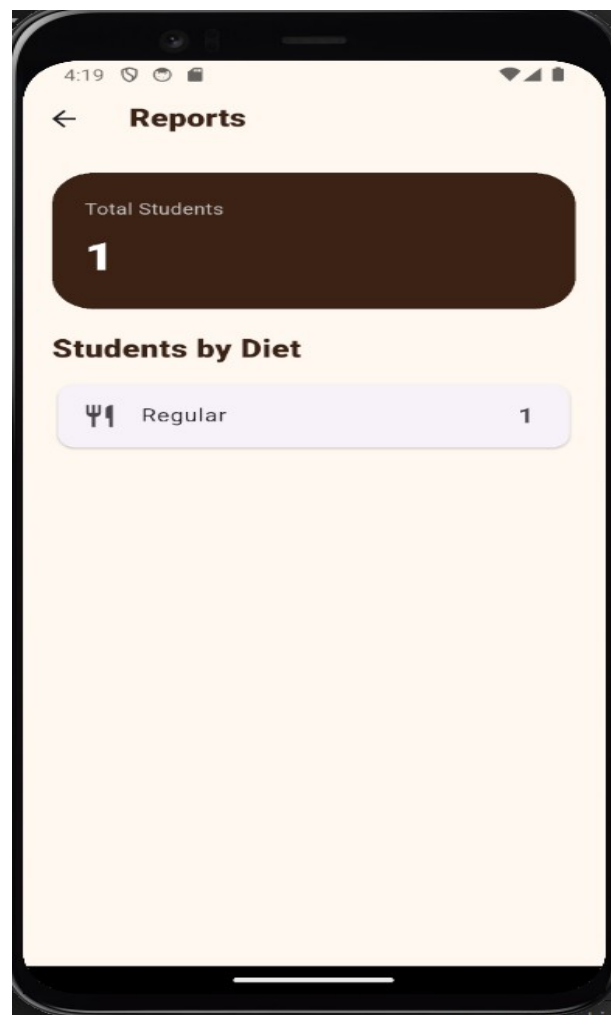
- Regular Diet
- Diabetic Diet
- Vegetarian Diet
- Peanut-Free Diet



Reports

The system generates reports showing:

- Total students
- Student distribution by diet category



Chapter 8: System Testing

Testing was conducted to evaluate the functionality and reliability of the application.

Functional Testing

- User Registration
- User Login
- Add Student
- Edit Student
- Delete Student
- Search Student
- Reports Generation

Performance Testing

- Application responsiveness
- Firebase connectivity
- Data loading speed

Results

All major system functions operated successfully.

Chapter 9: Benefits of the System

- Improves nutrition record management.
- Reduces manual paperwork.
- Enhances data accuracy.
- Improves accessibility of information.
- Supports nutrition monitoring.
- Provides real-time cloud storage.

Chapter 10: Challenges Encountered

During development, several challenges were encountered:

- Firebase configuration.
- Flutter project setup.
- Android emulator configuration.
- Authentication implementation.
- Cloud database integration.

These challenges were overcome through research, testing, and debugging.

Chapter 11: Future Enhancements

Future versions of NutriPlan Campus may include:

- Student profile photos.
- QR Code identification.
- PDF report generation.
- Notification system.
- Advanced analytics dashboard.
- Multi-user role management.

CHAPTER 12: TECHNOLOGIES USED

The following technologies were used during the development of NutriPlan Campus:

Technology	Purpose
Flutter	Mobile application development
Dart	Programming language used to build the application
Firebase Authentication	User registration and login management
Cloud Firestore	Cloud database for storing student information
Visual Studio Code	Development environment
GitHub	Version control and project hosting
Android SDK	Application testing and deployment

These technologies were selected because they provide a modern, reliable, and scalable platform for mobile application development.

Chapter 13: Conclusion

NutriPlan Campus successfully demonstrates the use of modern mobile technologies in solving student nutrition management challenges.

By integrating Flutter, Firebase Authentication, and Cloud Firestore, the system provides a secure, reliable, and scalable platform for managing student dietary information and nutrition records.

The project achieved its objectives and provides a strong foundation for future development and expansion.